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STEM Action Teen Institution

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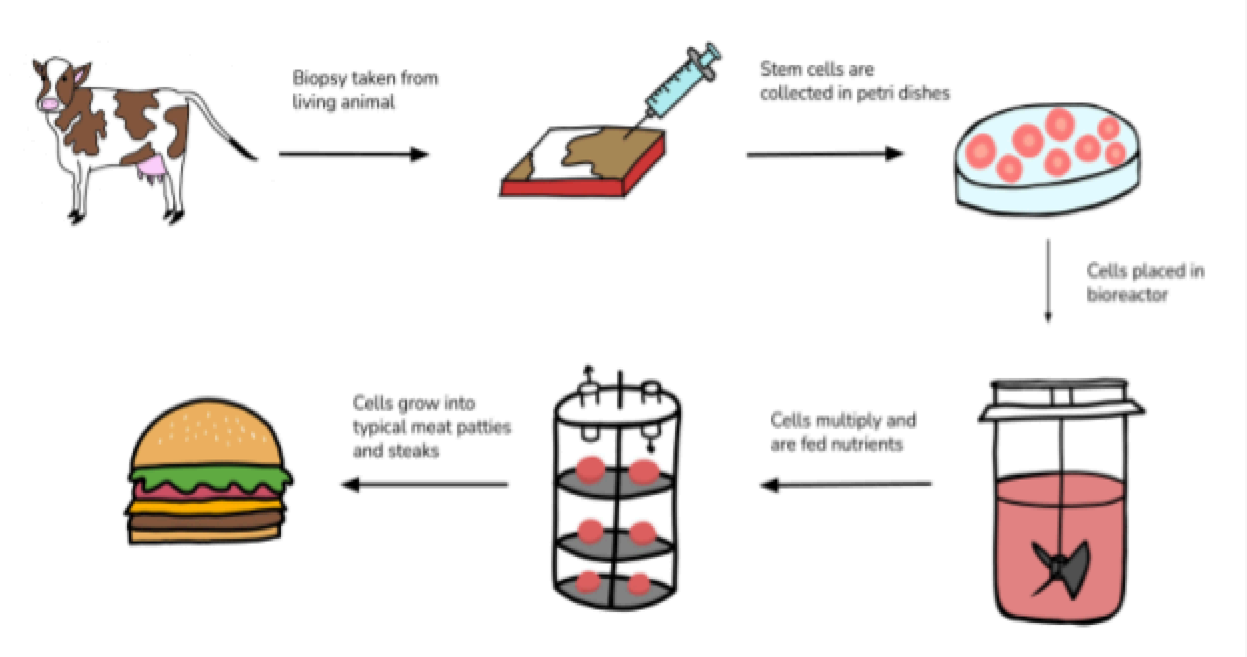
Lab-Grown Meat: How Cultivated Meat Could Revolutionize Food Production

Kevin Zhang

By 2050, when the world's population will be around 10 billion, there will be a dramatic rise in meat consumption. While conventional livestock farming meets this need, it now comes under fire for its negative impact on nature, morals, and the health of the general public. Growing meat from cells in a bioreactor provides a much-needed new option. Scientists can use healthy animal cells in laboratory conditions to produce meat-like muscle tissue, all without requiring animals. As a result, food production might be changed for the better by making safe and moral protein available.

Cultivated meat stands out mostly because it is eco-friendly. Usually, food grown on farms is resource-dependent, needing huge areas of land and water. And, it gives off significant amounts of greenhouse gases, including methane. In contrast, producing meat artificially could help cut greenhouse gas emissions by 96%, use 99% less land, and use 96% less water, reported the University of Oxford. In addition, by making meat from cells instead of animals, we remove the use of antibiotics which helps prevent antibiotic resistance in humans. This approach lowers the chance of diseases like swine flu or bird flu which are diseases that begin in animals, helping keep food everywhere safer. Such benefits make it possible that cultivated meat will be vital in lowering emissions and helping maintain a safe, sufficient supply of food.

Even so, there are still problems before lab-grown meat can be widely used. The technology is just beginning, so it takes considerable investment to increase production efficiently. Stiff prices, not enough awareness from the general public and barriers to clearance by regulators are slowing down its worldwide expansion. Besides, many consumers doubt the health and natural qualities of lab-grown meat which may stop them from using it. Yet, progress is happening faster: companies are refining how they produce meat, costs are falling and several nations have begun allowing cultivated meat to be sold. When supported by the scientific, government and commercial worlds, lab-grown meat has the chance to reshape the food industry, using an ethical and sustainable answer to a major challenge today.



A basic diagram of how cultivated meat is produced by Eliagrisanti
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Newton's Three Laws

Aidan Hong

If there is one thing constant at all times, it is Newton's Three Laws. Even though you might not notice it, it is always at work. Newton's three laws consist of three parts: inertia, F=ma, and that every reaction has an equal and opposite reaction.

The first law states that an object in motion stays in motion unless another force acts upon it. This explains why in space, things always move in one line – there is no other force acting upon the object. However, when we jump, we come back down and we don't keep going up. This is because there is another force acting upon us – gravity. While the force of us jumping is enough to overcome gravity for a brief moment, eventually, gravity overpowers the force of jumping and brings us back down.

The second law states that force is equivalent to mass times acceleration. This is a mathematical expression of force. Typically, when there are numerous forces acting on something, we get the net force, which is simply canceling out all the extra forces.

The third law states that every action has an equal and opposite reaction. This is another reason why, despite gravity acting on us, we don't get crushed to the ground. When you are standing, the force of gravity is acting on you, pushing you down. However, the ground is also pushing back up, negating the forces of gravity. There are also other examples of this as well, for example, dribbling a basketball. When you dribble a basketball, your hand pushes off the ball, and when the ball lands on the ground, the ground pushes back up on the basketball.

Newton's three laws are always in action. Although you may not realize it, it can be applied to anything at any moment. It also led to numerous other scientific advances related to physics. With this in mind, it is safe to say that Newton's Three Laws are one of the most fundamental and important discoveries of Physics.

Pulsars the Cosmic Lighthouse

Brandon Pian

Pulsars are like a lighthouse in space, they can be seen on Earth as a flickering star. They start as a massive star that turns into a Neutron star. Neutron stars form when a massive star runs out of fuel and collapses in on itself. The core of the star turns every proton and electron into a neutron. If the collapsing star has a solar mass of 1 to 3 the neutrons can stop the star from collapsing creating a neutron star. Most neutron stars can be seen as pulsars.

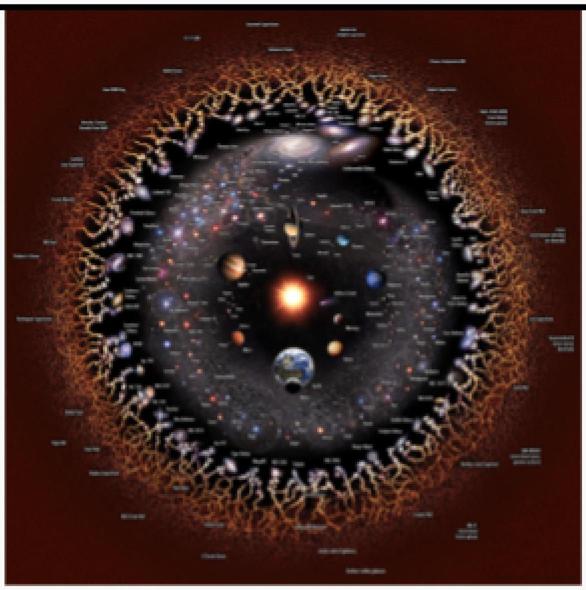
Pulsars are neutron stars that rotate and have pulses of radiation. Pulsars have strong magnetic fields that shoot jets of particles out along its two magnetic poles. The particles become very fast and produce powerful beams of light. Those beams are what makes Pulsars turn on and off. Pulsars allow scientists to learn more about the physics of neutron stars. This allows scientists to learn about the strange state of matter within neutron stars called nuclear pasta because of its shape. The precision of the pulses of pulsars is considered the most accurate natural clock. As a result, when there is a change in the pulsars blinking it could indicate something happening nearby.

If a pulsar formed from the wreckage of a Supernova it spins fast and emits a lot of energy. As pulsars age they rotate slower and they release less and less radiation. When pulsars slow down to a certain speed they die and enter the pulsar graveyard. If the dying pulsar is near a stellar composition it may take material and energy from the stellar composition to revive itself.

Is the Universe a Doughnut?

Yidian Wang

Have you ever wondered what the shape of the universe is, if it even has one? The hypothesis is that the universe is shaped like a doughnut; an Euclidean 3-torus. is a fascinating theory in cosmology. Many people may think the universe is infinite and ever-expanding with no end, and though this may be true, the universe being a doughnut would make more sense.



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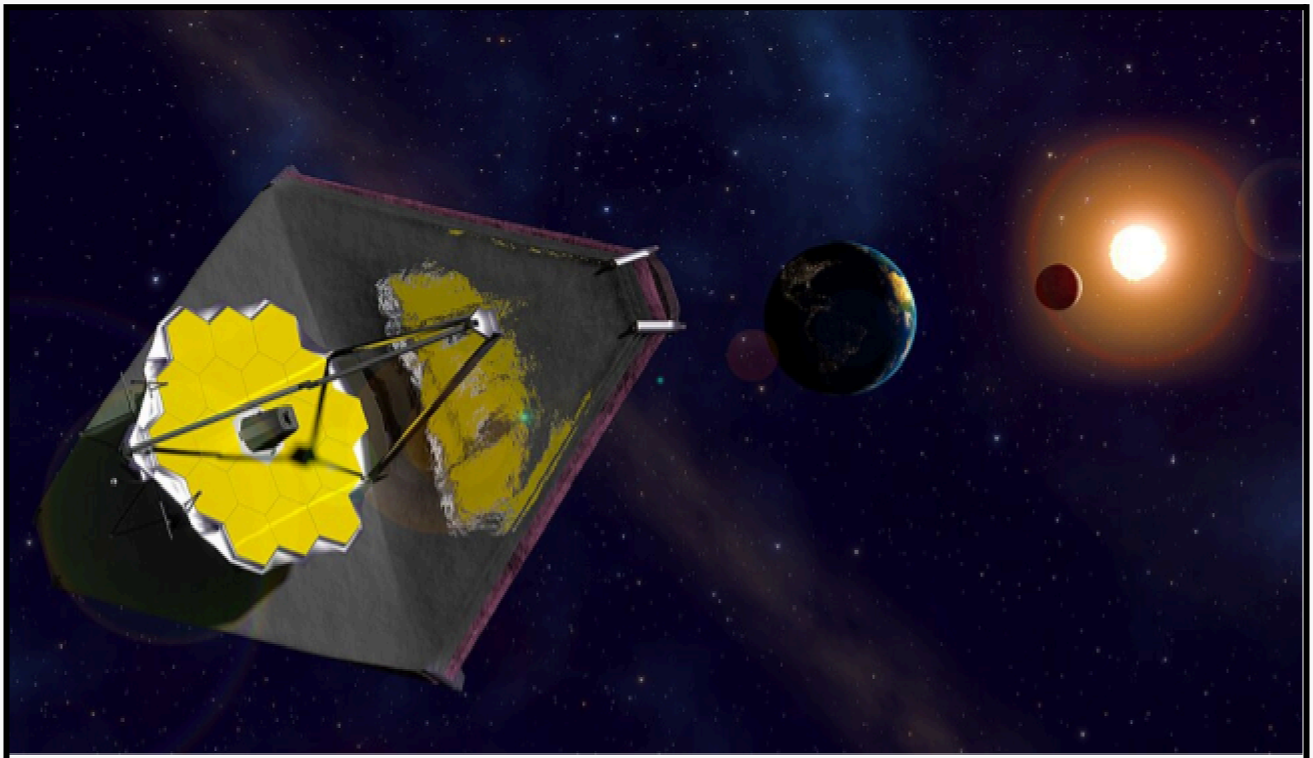
Doughnuts have a complex shape. You can travel across the entire object and end up back where you started, much like a sphere. Earth is in the shape of a sphere, so if you walk in a straight line from one point you will eventually travel the entire globe and end up in the same spot. The loop can be shrunk to a single point because it can be traveled in a closed loop. If the universe does have a doughnut shape, it means you could travel the whole cosmos and end up in the same spot. But unlike the Earth, you cannot squish it to a single point if you travel a loop around the hole because the hole gets in the way. This makes the doughnut a complex shape, or a “nontrivial topology.”

Scientists are still looking for signs of the universe looping back on itself. They've looked at the cosmic microwave background, essentially searching for identical shapes of light that appear in different places in the sky, and for similarities between different areas. Although there's still no proof that the universe is in the shape of a doughnut, the boundless complexity of the universe and our minimal knowledge of it allow for more improvements and discoveries to emerge in the future.

James Webb Telescope

Richard Wang

This telescope, more refined than the Hubble, can see objects as old as glows after the big bang. First launched on December 25, 2021, the James Webb Space Telescope (JWST) has already delivered the sharpest infrared image of the distant universe and is capable of tracing galaxies to the beginning of their time.



Rendered in the existing scene I had for the Earth and Mars. Using Blender 2.71 with Cycles. Post processing in Photoshop.by Kevin Gill from WikiMedia Commons
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The telescope utilizes infrared light, unable to be perceived by the human eye. Objects around us give off radiation, and those varieties of radiation are found in the em (electromagnetic) spectrum. However, we can only see a small part of radiation which is called visible light. Infrared light when compared to visible light is longer, which allows it to slip more easily in between particles. Stars and planetary systems covered in clouds of gas are opaque to visible light but with infrared vision gives us the ability to now see through them.

Some main components of the JWST include its sunshield with a size equivalent to a tennis court. There is also a cooling system in order to keep extremely cold temperatures as heat from the sun will interfere with observations. The telescope also sends transmitter signals that are received by countless, large antennas dispersed around the world. There is a gold mirror made up of 18 hexagonal gold segments. It reflects high percentages of infrared light and is used because gold is relatively unreactive which means it won't tarnish easily.

A mission that is being aimed to be achieved by the JWST is observing deeper into the universe than ever before. It also aims to discover the first stars after the big bang. It hopes to better understand the birthhood of stars and galaxies and how they develop. Most importantly, it focuses on determining potential life on other planetary objects. There is more to be discovered by the JWST and the possibilities are endless.

Foreign Accent Syndrome

Denise Lee

Foreign Accent Syndrome (FAS) is a rare and fascinating neurological disorder in which a person suddenly begins speaking with a perceived foreign accent, usually following brain injury or trauma. People with Foreign Accent Syndrome (FAS) begin to sound as if they are speaking a foreign language, and this often occurs as a result of a brain injury or accident. This condition doesn't happen because the person plans to change their words. Even so, it occurs when injuries to the speech and motor control parts of the brain occur, for example, in the left hemisphere's speech zones.

The first studies of FAS in 1907 showed it was connected to strokes, traumatic brain injuries, and some mental illnesses. People affected by FAS will continue to use their own language and vocabulary, though there is a noticeable change in how they say things, so they seem to have an accent. An American English speaker might suddenly talk with a British or French accent, even without ever being there.

Only a tiny number of cases of FAS have been recorded all over the world. Usually, doctors perform brain scans, analyze your speech, and do neurological checks to make sure aphasia or speech apraxia is not the cause. Patients may often feel emotional and social distress because of the condition. A lot of people feel lonely or fail to relate to, mostly when others falsely believe they are changing their accent. People may be helped with speech therapy, counseling, and joining support groups to adapt and gain confidence in their speaking.

The condition is still an interesting mystery, revealing the important link between how the brain and speech work. More study could explain the processes involved in generating and controlling language by our brains.

The Significance of Electric Potential

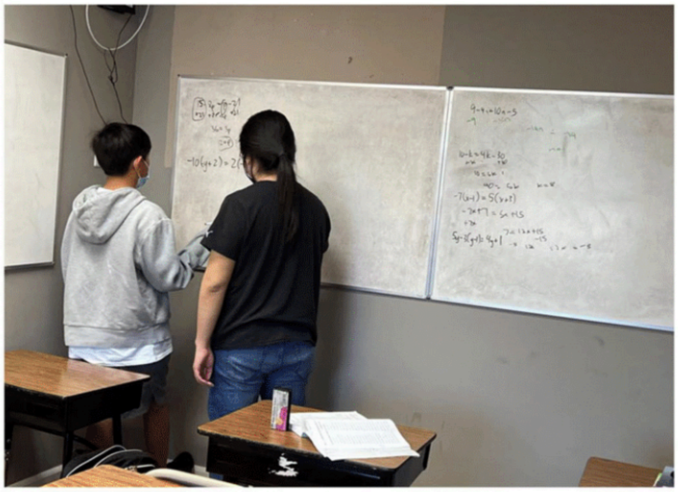
Kenny Wu

Faraday's Cage is a prominent concept in electromagnetism, reflecting the ability of conductive materials to shield interior spaces from external electric fields. The cage is named after the English scientist Michael Faraday, who first demonstrated its effect in 1836, this principle reveals how electric charges on a conductor redistribute to cancel incoming fields, ensuring the space within remains unaffected. This phenomenon has become essential not only to theoretical physics but also countless practical applications in modern technology.

One of the most direct uses of a Faraday cage is in protecting sensitive electronics from electromagnetic interference (EMI). Laboratory instruments, satellites, and military equipment all employ conductive enclosures to preserve the accuracy and function of their systems. The cage effectively acts as a barrier, allowing normal operation even in the presence of disruptive external fields.

In medicine, entire rooms are sometimes designed as Faraday cages to ensure the precision of machines like MRI scanners. These rooms prevent outside radio frequencies from distorting internal imaging, allowing doctors to diagnose patients accurately. The same principle is applied in everyday appliances, microwave ovens trap radiation within their metal walls to cook food efficiently and safely.

Ultimately, Faraday's Cage demonstrates how a straightforward scientific idea can have vast implications. Its utility in shielding against unwanted electric fields helps preserve the stability of complex systems and highlights the power of physics to bring both clarity and protection in an increasingly electrified world. Understanding this concept not only deepens one's appreciation of electromagnetic theory but also reveals its silent presence in the safety and precision of modern life.



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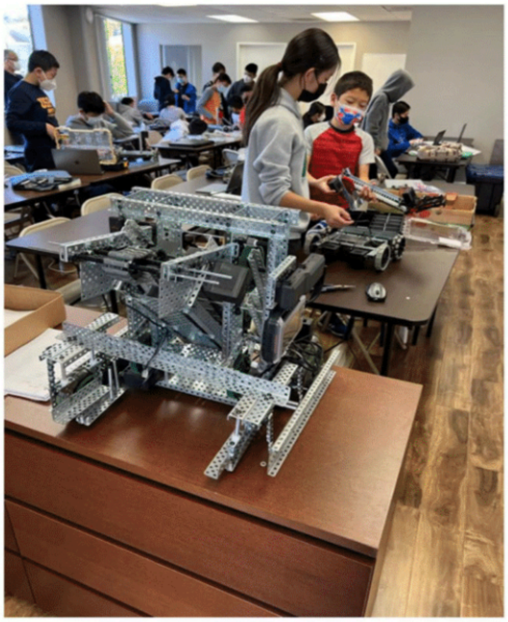

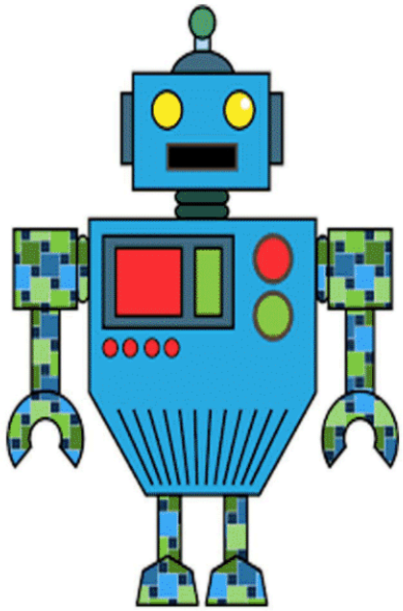

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